Raffles Institution Physics Department Revision Package 1	# Thermodynamic Systems	13

Important note: **denotes questions that may require knowledge from later chapters in order to solve them.*

I. MCQs

1. **[AJC/H2/Prelims 2010/P1/15]**

The table summarises some properties of evaporation.

Which row of the table is correct?

	occurring at a fixed temperature	involving a reduction in the average kinetic energy of the remaining atoms
A	true	true
B	true	false
C	false	false
D	false	true

2. **[AJC/H2/Prelims 2010/P1/16]**

What is the internal energy of an object?

A It is the total energy associated with the object's movement and position in space.

B It is the energy associated with the random movement of the molecules in the object.

C It is the energy due to the attractions between the molecules within the object.

D It is the sum of all the microscopic potential and kinetic energies of the molecules of the object.

3. **[CJC/H2/Prelims 2010/P1/18]**

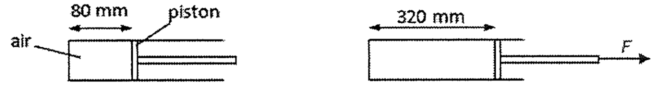
A metal block **X**, of mass *m*, at 0°C comes into contact with another metal block **Y**, of mass 2*m*, at 100°C. Heat conduction takes place with no loss to the surroundings. The final equilibrium temperature of the blocks is 20°C.

If the specific heat capacities of the two metals are c_X and c_Y respectively, then

A $c_X = 8c_Y$ B $c_X = 4c_Y$ C $c_X = 2c_Y$ D $c_X = ½c_Y$

4. **[CJC/H2/Prelims 2010/P1/19]**

Air is enclosed in a cylinder by a gas-tight, frictionless piston of cross-sectional area 3.0×10^{-3} m^2. When atmospheric pressure is 100 kPa, the piston settles 80 mm from the end of the cylinder. The piston is then pulled out until it is 320 mm from the end of the cylinder and is held there. The temperature of the air in the cylinder returns to its original value.



What is the force *F* required to hold the piston in its new position?

A 75 N B 100 N C 225 N D 300 N

5. **[CJC/H2/Prelims 2010/P1/20]**

Which statement is **not** correct?

A Boiling always occurs at a higher temperature than evaporation.

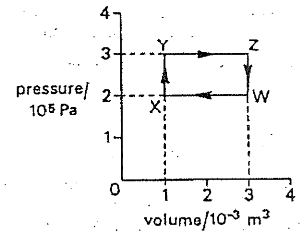
B In boiling, the most energetic molecules of the liquid escape leaving behind molecules with lower energies, while in evaporation, all molecules have the same energy.

C Evaporation occurs at any temperature but the boiling point depends on the external pressure.

D The rate of evaporation is dependent on the surface area of the liquid.

6. **[IJC/H2/Prelims 2010/P1/17]**

A gas undergoes the cycle of pressure and volume changes W → X → Y → Z → W as shown in the diagram.



What is the net work done on the gas?

A −600 J B −200 J C 0 J D 200 J

7. **[MI/H2/Prelims 2010/P1/17]**

Which statement is true?

A The internal energy of a system can be increased by transfer of energy by heating.

B The internal energy of a system is dependent only on its temperature.

C When the internal energy of a system is increased, its temperature always rises.

D When two systems have the same internal energy, they must be at the same temperature.

8. **[MI/H2/Prelims 2010/P1/18]**

A heater of power of 300 W is immersed in a filter funnel of crushed ice. Before the heater is switched on, 10 g of water is collected from the melting ice in 1.0 minute. When the heater is switched on, the mass of water collected from the melting ice in 1 minute is 85 g.

What is the specific latent heat of fusion of ice, in J g^{-1}?

A 5.0 B 240 C 300 D 1200

9. **[MJC/H2/Prelims 2010/P1/21]**

Which statement about internal energy is correct?

A The internal energy of a system can be increased without transfer of energy by heating.

B The internal energy of a system is the sum of the kinetic energies of the molecules.

C When the internal energy of a system is increased, its temperature always rises.

D When two systems have the same internal energy, they must be at the same temperature.

10. **[NJC/H2/Prelims 2010/P1/20]**

A water tank of heat capacity 5000 J K^{-1} contains 10 kg of water at 25°C.

What is the time taken to raise the temperature of the water to 45°C using a heater coil of power of 3.0 kW?

The specific heat capacity of water is 4200 J kg^{-1} K^{-1}.

A 61 s B 280 s C 310 s D 610 s

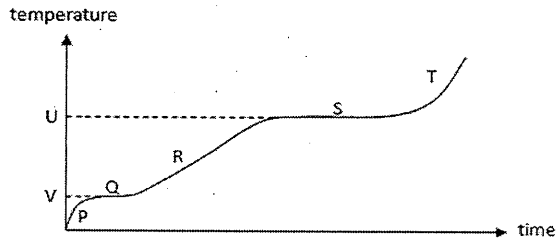
11. **[NYJC/H2/Prelims 2010/P1/18]**

An ideal monatomic gas has 1000 J of heat added to it and it does 500 J of work; its temperature changes by T_1. When twice the amount of heat is added to it and it does the same amount of work, its temperature changes by T_2. What is the ratio of T_1/T_2?

A 1/5 B 1/3 C 3/5 D 1

12. **[PJC/H2/Prelims 2010/P1/17]**

A container of ice is heated by an electric heater. The graph below shows the variation of the temperature of the ice with time.



Which part of the graph shows that the specific latent heat of vaporisation of water is greater than its specific latent heat of fusion?

A The gradient of the graph at T is greater than the gradient at R.

B The length of line S is greater than the length of line Q.

C The gradient of the graph at P is greater than the gradient at R.

D The value of U is greater than the value of V.

13. **[RI/H2/Prelims 2010/P1/17]**

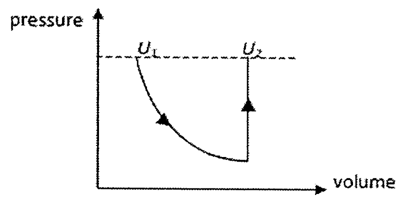
Four different solids A, B, C and D of equal masses at 20°C are separately heated at the same rate. Their melting points and specific heat capacities are as shown in the table below.

Which of these solids will start to melt first?

	melting point / °C	specific heat capacity / J kg^{-1} K^{-1}
A	80	1200
B	100	800
C	150	600
D	300	250

14. **[RVHS/H2/Prelims 2010/P1/18]**

A sample of an ideal gas initially having internal energy U_1 is allowed to expand under constant temperature by absorbing heat Q_1 and doing external work W. Heat Q_2 is then supplied to it, keeping the volume constant at its new value, until the pressure returns to its original value. The final internal energy is U_2.

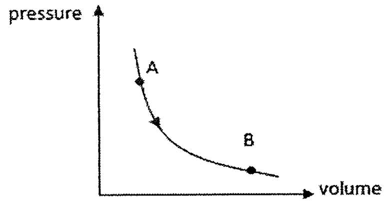


The increase in internal energy, $U_2 - U_1$, is equal to

A 0 B $Q_1 + Q_2$ C $W + Q_2$ D Q_2

15. [SAJC/H2/Prelims 2010/P1/16]

In the figure below, the curve is an isotherm (a curve which joins up all the points having the same temperature) for a fixed mass of ideal gas.



Which of the following statements can be deduced for the process from A to B?

A Positive work is done on the gas and heat is supplied to the gas.

B Positive work is done by the gas without any heat supplied to the gas.

C The internal energy of the gas decreases as heat is released by the gas.

D Heat is supplied to the gas and the internal energy of the gas remains unchanged.

16. [TJC/H2/Prelims 2010/P1/18]

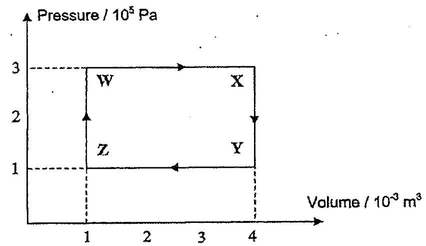
A solar furnace made from a concave mirror of area 0.40 m^2 is used to heat water. The radiant energy from the sun arrives at the mirror's surface at a rate of 1400 W m^{-2}. The specific heat capacity of water is 4200 J kg^{-1} K^{-1}.

What is the best estimate of the least time that the furnace takes to heat 2.0 kg of water from 20°C to 50°C?

A 3.8 min B 7.5 min C 230 min D 450 min

17. [SAJC/H2/Prelims 2010/P1/17]

An ideal gas undergoes the cycle of pressure and volume changes W → X → Y → Z as shown in the diagram.

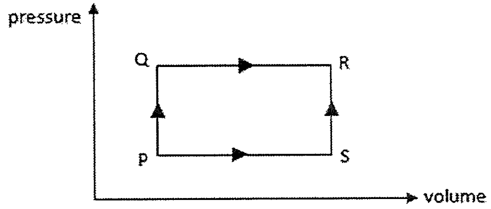


What is the work done by the gas in the process Y → Z and the net work done by the gas as it undergoes a complete cycle of pressure and volume change?

	work done by the gas in process Y → Z	net work done by the gas
A	–300 J	600 J
B	300 J	–600 J
C	–400 J	1200 J
D	400 J	–1200 J

18. [TJC/H2/Prelims 2010/P1/20]

A fixed mass of ideal gas undergoes changes of pressure and volume as shown.



When the gas is taken from state P to state R by the stages PQ and QR, 8 J of heat are absorbed by it and 3 J of work are done by it. When the same resultant change is achieved by stages PS and SR, 1 J of work is done by the gas.

In this case,

A 12 J of heat are liberated.

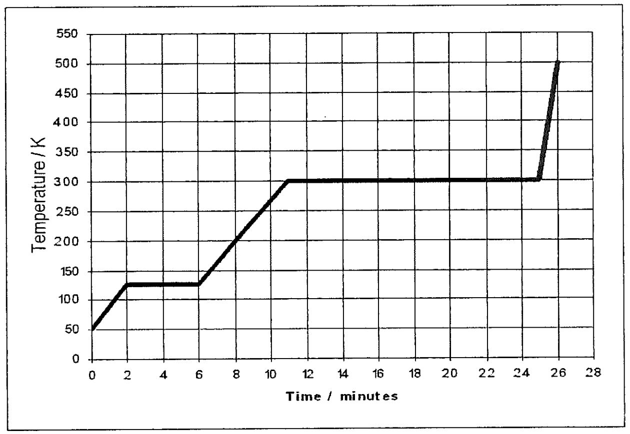
B 10 J of heat are absorbed.

C 8 J of heat are absorbed.

D 6 J of heat are absorbed.

19. [VJC/H2/Prelims 2010/P1/17]

In the figure below, heat is added to a pure substance in a closed container at a constant rate. A graph of the temperature of the substance as a function of time is shown.



L_f is the latent heat of fusion and L_v is the latent heat of vaporization of the substance.

What is the value of the ratio $\frac{L_v}{L_f}$ for this substance?

A 1.0 B 2.4 C 3.5 D 5.0

II. Structured Questions

1. **[AJC/H2/Prelims 2010/P2/3]**

A fixed mass of gas in a heat pump undergoes a cycle of changes of pressure, volume and temperature as illustrated in Fig. 1.1. The gas is assumed to be ideal.

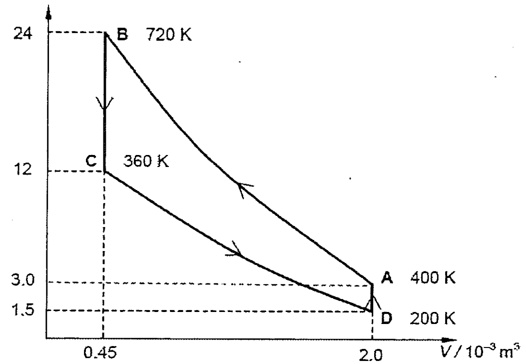


Fig. 1.1

a. Determine the number of moles of the gas. **[1]**

b. Complete the table below indicating the energy changes in each stage of the cycle. **[4]**

	Increase in internal energy / J	Heat supplied to gas / J	Work done on gas / J
A to B	720	0	
B to C	–810		
C to D	–360	0	
D to A			

c. State one practical use of such a heat pump. **[1]**

[Total: 6]

2. **[CJC/H2/Prelims 2010/P3/2]**

a. State what is meant by the *internal energy* of a system. **[2]**

b. An ideal gas in a cylinder can be considered to undergo a cycle of changes of pressure, volume and temperature as shown on the graph of Fig. 2.1

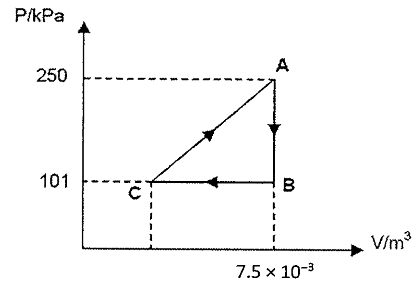


Fig. 2.1

The temperature of the gas at A and C are 623 K and 50 K respectively.

i. Calculate the number of gas molecules in the cylinder. **[2]**

ii. Calculate the volume of gas at C. **[2]**

iii. Calculate the net work done by the gas. **[2]**

iv. State with a reason the total change in internal energy of the gas when it completes a cycle. **[2]**

[Total: 10]

3. **[CJC/H2/Prelims 2010/P2/4 (part)]**

A student sets up the apparatus illustrated in Fig. 3.1 in order to determine a value for the specific latent heat of fusion of ice.

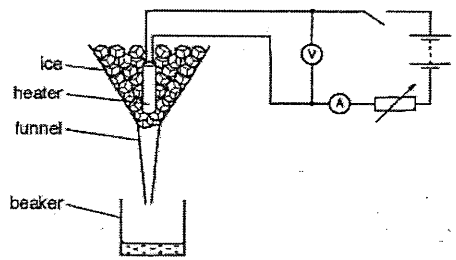


Fig. 3.1

A heater is placed in the funnel, surrounded by pure melting ice. The student measures the mass of melted ice in the beaker at regular time intervals before and after switching on the heater. The variation with time t of the mass m of melted ice in the beaker is shown in Fig. 3.2.

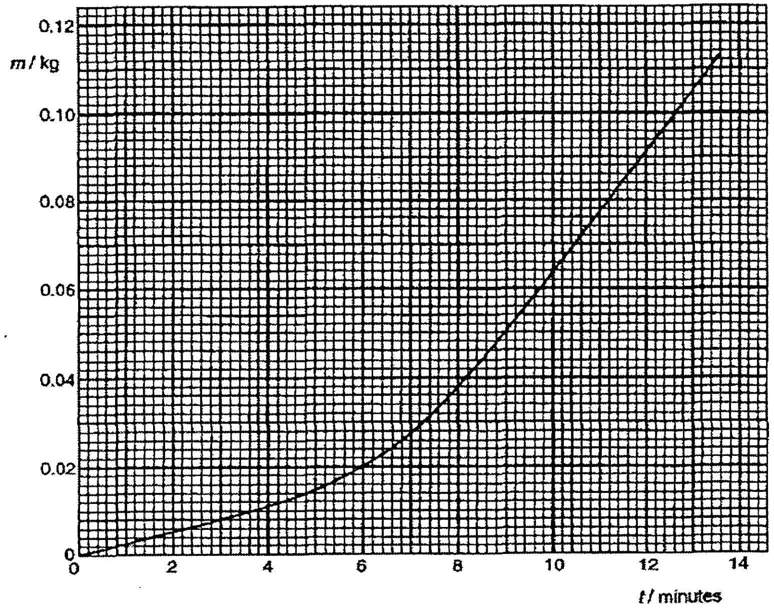


Fig. 3.2

During the heating process, the current is adjusted so that the readings on the ammeter and voltmeter are constant. By reference to Fig. 3.2,

a. State

i. suggest a time at which the heater is switched on; [1]

ii. determine the mass of ice melted in 1.0 minute

1. with the heater switched off, [1]

2. with the heater switched on. [1]

b. The readings of the ammeter and the voltmeter are 5.2 A and 11.5 V respectively. Use your answers in **a.** to calculate a value for the specific latent heat of fusion of ice. [3]

[Total: 6]

4. [Dunman High/H2/Prelims 2010/P3/3]

Ice of 50 g at temperature –15°C is added to water at 30°C in an insulated container.

Specific heat capacity of ice = 2100 J kg^{-1} K^{-1}

Specific heat capacity of water = 4200 J kg^{-1} K^{-1}

Specific latent heat of fusion of water = 3.3×10^5 J kg^{-1}

a. Calculate the quantity of energy needed to change the ice at –15°C to water at 0°C. [3]

b. Calculate the mass of the water in the container if the lowest temperature reached by the water is 7.5°C, assuming no heat is lost to the surroundings. [3]

c. State one other assumption that you have made in your calculations in **b.**. [1]

[Total: 7]

5. [IJC/H2/Prelims 2010/P3/2]

a. i. What is meant by the term *internal energy of a system*? [2]

ii. Write down an equation representing first law of thermodynamics. Define the symbols that you use. [2]

b. Some water in a saucepan is boiling.

i. Explain why there is a change in internal energy as water changes to steam. [2]

ii. Explain why work is done by the boiling water. [2]

iii. With reference to your answers in **b.i.** and **b.ii.**, show that thermal energy must be supplied to the water during the boiling process. [2]

[Total: 10]

6. **[HCI/H2/Prelims 2010/P3/2]**

   a. State

      i. the *first law of thermodynamics;* **[2]**

      ii. the meaning of the term *internal energy.* **[2]**

   b. The diesel cycle is the thermodynamic cycle which approximates the pressure and volume of the combustion chamber of the diesel engine, invented by Rudolph Diesel in 1897.

      An ideal gas undergoes the diesel cycle which comprises 4 processes (Fig. 6.1):

      Process **A** - adiabatic compression
      Process **B** - isobaric heating
      Process **C** - adiabatic expansion
      Process **D** - isovolumetric cooling

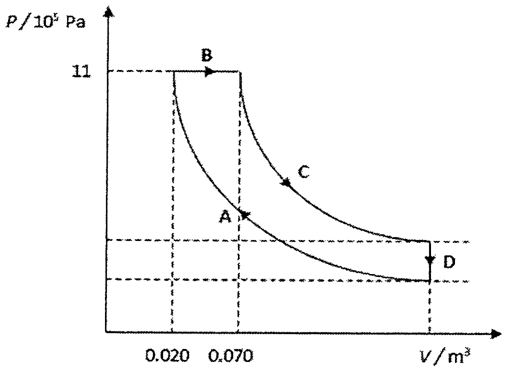


Fig. 6.1

   i. Calculate the work done by the gas during process **B**. **[2]**

   ii. Table 6.1 is a table of energy changes during one cycle. Complete the table with appropriate values. **[4]**

Process	work done on gas, W / kJ	heat supplied to gas, Q / kJ	increase in internal energy, ΔU / kJ
A	+201		
B		+74	
C	−185		
D		−35	−35

Table 6.1

[Total: 10]

7. **[MI/H2/Prelims 2010/P2/3 (part)]**

   a. Using the simple Kinetic Model of Matter, explain the following:

      i. The melting of ice takes place without a change in temperature. **[2]**

      ii. The specific latent heat of vaporisation of water is higher than its specific latent heat of fusion. **[2]**

[Total: 4]

8. **[MJC/H2/Prelims 2010/P3/5]**

   a. Explain what is meant by the *internal energy* of a system. **[2]**

   b. State the process and give one practical example of each of the following:

      i. a process in which heat is supplied to a system without causing an increase in temperature, **[2]**

      ii. a process in which no heat enters or leaves a system but the temperature changes. **[2]**

[Total: 6]

9. **[NJC/H2/Prelims 2010/P3/2]**

   A monoatomic ideal gas is subject to a cycle of changes **ABCA**. Fig. 9.1 shows a graph of pressure p against volume V for one cycle of changes for the gas.

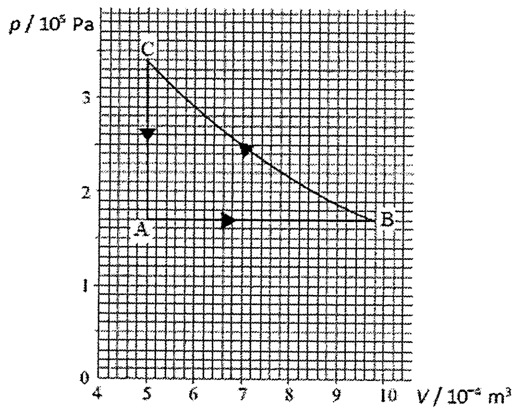


Fig. 9.1

   a. i. Using data from the graph, verify that process BC is isothermal. Show your workings clearly. **[2]**

      ii. Explain the term *internal energy* in relation to an ideal gas. **[1]**

   b. Temperature of the gas at point C is 385 K. Calculate the temperature of the gas at point A in °C. **[1]**

   c. i. Calculate the change in the internal energy of the gas during the process **AB**. **[2]**

      ii. Work is done by the gas in the change **AB**. State what must be done to the system for this change to occur. Explain using the first law of thermodynamics. **[2]**

[Total: 8]

10. [NYJC/H2/Prelims 2010/P3/7 (modified)]

a. Explain what is meant by the *internal energy* of a system. [2]

b. A cake of mass 0.90 kg is cooked in an oven at a temperature of 180°C. It is taken out of the baking tin onto a rack to cool in a kitchen of 20°C.

i. State the final absolute temperature of the cake. [1]

ii. Calculate the energy released from the cake in cooling.
The specific heat capacity of the cake is 990 J kg^{-1} K^{-1}. [1]

c. The oven of volume 0.10 m^3 cools down from 180°C to 25°C.
Calculate the change in the mass Δm of air in the oven between the two temperatures.
The pressure in the oven remains at an atmospheric pressure of 1.0×10^5 Pa. Assume that air behaves ideally.
[Relative molecular mass of air = 0.030 kg mol^{-1}] [4]

d. The gas in the cylinder of a diesel engine can be considered to undergo a cycle of changes of pressure, volume and temperature. One such cycle, for an ideal gas, is shown in Fig. 10.1.

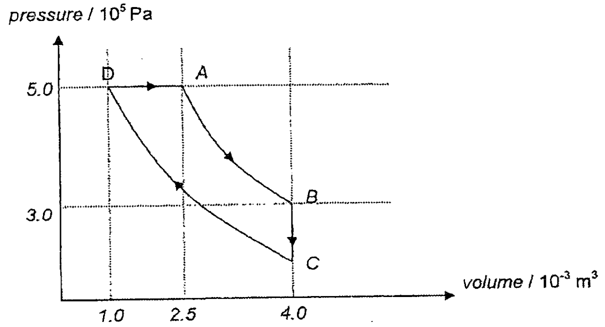


Fig. 10.1

The table below shows the increase in internal energy which takes place during each of the changes A to B, B to C, C to D and D to A.

	Heat supplied to gas / J	Work done on gas / J	Increase in internal energy of gas / J
A to B	0	–300	–300
B to C	–450	0	–450
C to D	0	650	650
D to A			

Using Fig. 10.1, fill in the missing values in the table above. [3]

e. In a continuous flow method for determining the specific heat capacity a liquid, the liquid flows through the tube at 0.15 kg min^{-1}, while the heater provides power at 25 W. The temperatures of the liquid at the inlet and outlet are 15°C and 19°C, respectively.
With the inlet and outlet temperatures unchanged, the flow rate is increased to 0.23 kg min^{-1} and the power of the heater is increased to 37 W.

i. Explain why it is necessary for the inlet and outlet temperatures to remain unchanged. [1]

ii. Determine the rate of heat loss of the liquid. [3]

[Total: 16]

11. [RI/H2/Prelims 2010/P2/4]

a. Explain what is meant by the *internal energy* of a gas. [1]

b. A cylinder fitted with a piston contains 0.20 moles of an ideal gas. Initially the volume and pressure of the gas are 5.0×10^{-3} m^3 and 1.0×10^5 Pa respectively.

i. Calculate the initial temperature of the gas. [2]

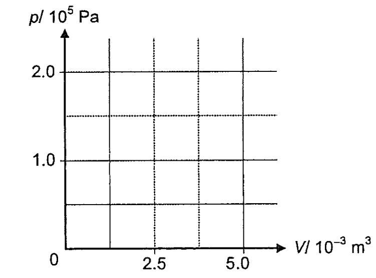
ii. The gas is

(1) heated at constant volume to twice its initial temperature

(2) cooled at constant pressure to its initial temperature, and finally

(3) expanded isothermally to its initial volume.

Sketch the above changes on a clearly labelled *p-V* diagram. [4]



[Total: 7]

12. [PJC/H2/Prelims 2010/P3/6]

a. A car tyre has a fixed internal volume of 0.0160 m^3. On a day when the temperature is 27°C, the pressure in the tyre has to be increased from 2.76×10^5 Pa to 3.91×10^5 Pa.

i. Assuming that the air is an ideal gas, calculate the amount of air which has to be supplied at constant temperature. [3]

ii. A portable supply of air used to inflate tyres has a volume of 0.0117 m^3 and is filled with air at a pressure of 1.165×10^6 Pa. Show that, at 27°C, there is more than enough air in it to supply four tyres, as in a.i., without the pressure falling below 4.00×10^5 Pa. [3]

iii. Show that the average kinetic energy of a molecule of air at a temperature of 27°C is 6.21×10^{-21} J.
Assume that the air behaves as a monatomic ideal gas. [2]

iv. Hence, calculate the internal energy of one mole of the air at a temperature of 27°C. [1]

b. In order to study the sudden compression of a gas, some dry air is enclosed in a cylinder fitted with a piston, as shown in Fig. 12.1.

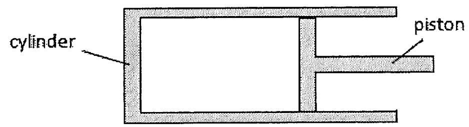


Fig. 12.1

The mass of air in the cylinder is constant. The material of the cylinder and the piston is an insulator so that no thermal energy enters or leaves the air.

The volume and pressure of air are measured. The piston is then moved suddenly to compress the air and the new volume and pressure are measured.

The variation with volume V of the pressure p of the air is shown in Fig. 12.2.

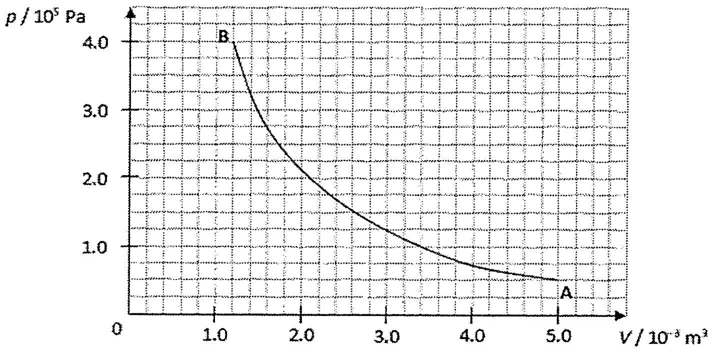


Fig. 12.2

It may be assumed that the dry air behaves as an ideal gas.

i. By considering the pressure and volume of the dry air at points A and B, and using the equation of state for an ideal gas, show that the temperature of the air increases when the air is compressed. [3]

ii. The dry air then goes through two more processes.

Process 1: The gas is cooled while keeping the piston at the same position.

Process 2: The gas then expands, while being kept at constant temperature, to return to its original state.

On Fig. 12.2, draw and label the p-V graphs of the two processes described above. [3]

[Total: 15]

13. [SAJC/H2/Prelims 2010/P3/6]

a. There is no attraction between the molecules of an ideal gas. Use this information to explain why the internal energy of an ideal gas is proportional to its temperature. [3]

b. Hence, explain how this relationship between the internal energy of an ideal gas and the absolute temperature gives rise to the concept of an absolute zero of temperature. [2]

[Total: 5]

14. [TJC/H2/Prelims 2010/P2/4]

a. State, in words, the relation between the increase in internal energy of a gas, the work done by the gas, and the heat supplied to the gas. [1]

[Total: 1]

15. [YJC/H2/Prelims 2010/P3/2 (part)]

a. Fig. 15 shows data for ethanol.

Density	0.79 g cm^{-3}
Specific heat capacity of liquid ethanol	2.4 J g^{-1} K^{-1}
Specific latent heat of fusion	110 J g^{-1}
Specific latent heat of vaporisation	840 J g^{-1}
Melting point	−120°C
Boiling point	78°C

Fig. 15

Use the data in Fig. 15 to calculate the thermal energy required to convert 1.0 cm^3 of ethanol at 20°C into vapour at its normal boiling point. [3]

b. Suggest why there is a considerable difference in magnitude between its specific latent heat of fusion and vaporization. [1]

[Total: 4]

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III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
D	D	A	C	B	B	A	B	A	C	B	B	B	D	D

16	17	18	19
B	A	D	C

1 **D** Evaporation occurs at all temperatures. During evaporation, the more energetic molecules at the surface of the liquid overcomes the forces of attraction and turns to gas state.

2 **A** The internal energy of a system is determined by the state of the system and it is the sum of the random distribution of kinetic and potential energies associated with the molecules of the system.

3 **D** Since all the thermal energy loss by Y is gained by X,

$$Q = mc\Delta\theta$$
$$|Q_X| = |Q_Y|$$
$$mc_X(20) = 2mc_Y(80)$$
$$c_X = 8c_Y$$

4 **C**

$$P = \frac{nRT}{V} \Rightarrow P \propto \frac{1}{V}$$

$P_i = 100$ kPa

since volume quadrupled, $P_f = 25$ kPa

$$P = \frac{F}{A}$$
$$F = 75000 \times 3.0 \times 10^{-3} = 225 \text{ N}$$

5 **B** Boiling occurs in the whole liquid and not only at the surface of liquid.

During evaporation, only the most energetic molecules escape.

6 **B** Area under P-V graph gives work done

$$1 \times 10^5 \times (3-1) \times 10^{-3} = 200 \text{ J}$$

Since the cycle is clockwise, work is done by the gas in the cyclic process and work done on gas is –200 J.

7 **A** Using 1st Law of thermodynamics, this statement is true.

8 **B** Since 10 g of water melts in 1.0 minute due to heat gain from surroundings,

$$Q = mL$$
$$300 \times 60 = 75 \times L$$
$$L = 240 \text{ J g}^{-1}$$

9 **A** The internal energy of a system is determined by the state of the system and it is the sum of the random distribution of kinetic and potential energies associated with the molecules of the system.

10 **C**

$$Q_{water} = mc\Delta\theta = 10 \times 4200 \times (45-25)$$
$$Q_{tank} = 5000 \times (45-25)$$
$$P = \frac{E}{t} \Rightarrow t = \frac{Q_{total}}{3000} = \frac{940000}{3000} = 313 \text{ s}$$

11 **B**

$$\Delta U = Q + W$$
$$\frac{3}{2}KT_1 = 1000 + (-500) = 500$$
$$\frac{3}{2}KT_2 = 2000 + (-500) = 1500$$
$$\frac{\frac{3}{2}KT_1}{\frac{3}{2}KT_2} = \frac{T_1}{T_2} = \frac{1}{3}$$

12 **B** The short horizontal line at Q means that little thermal energy is required for melting. The long horizontal line at S means that more thermal energy is required for boiling.

Note: The steeper the gradient of *T-t* graph, the less time is required for the system to increase temperature by a fixed amount and thus the **lower** the specific heat capacity of system.

13 **B**

$$Q = mc\Delta\theta$$
$$Q_A = m(1200)(80-20) = 72000m$$
$$Q_B = m(800)(100-20) = 64000m$$
$$Q_C = m(600)(150-20) = 78000m$$
$$Q_D = m(250)(300-20) = 70000m$$

14 D $\Delta U = Q + W$

process 1:

$0 = Q_1 + (-W)$

process 2:

$\Delta U = Q_2 + 0$

$U_2 - U_1 = Q_2$

15 D Since process A to B lies on an isotherm, temperature is kept constant throughout the process and thus no change in internal energy of system during the entire process.

$\Delta U = Q + W$

$0 = Q + W$

From A to B, the gas expands and thus positive work is done by gas and heat is supplied to gas.

16 B $Q = mc\Delta\theta$

$2.0 \times 4200 \times 30 = 1400 \times 0.40 \times t$

$t = 450 \text{ s} = 7.5 \text{ min}$

17 A net work done by gas = area WXYZ $= (3-1)\times 10^5 \times (4-1)\times 10^{-3} = 600$ J

work done by gas from Y to Z is negative and equal to area under PV graph

$= 1\times 10^5 \times (4-1)\times 10^{-3}$

$= -300$ J

18 D using 1st Law of thermodynamics, $\Delta U = Q + W$

$P \to Q \to R$: $\Delta U = 8 + (-3) = 5$ J

$P \to S \to R$: $\Delta U = 5$ J (since final states P and R are the same)

$5 = Q + (-1)$

$Q = 6$ J

19 C $Q = mL$

$P \times 4 = mL_f$

$P \times 14 = mL_v$

$\dfrac{L_v}{L_f} = \dfrac{14}{4} = 3.5$

Structured Questions

1. [AJC/H2/Prelims 2010/P2/3]

a. By the ideal gas equation, at point D:

$pV = nRT$

$$n = \frac{pV}{RT} = \frac{(1.5\times10^5)\times(2.0\times10^{-3})}{(8.314)\times(200)} = 0.180 \text{ mol}$$

b.

	Increase in internal energy / J	Heat supplied to gas / J	Work done on gas / J
A to B	720	0	**720**
B to C	–810	**–810**	**0**
C to D	–360	0	**–360**
D to A	**450**	**450**	**0**

c. Air conditioner / refrigerator

2. [CJC/H2/Prelims 2010/P3/2]

a. See definitions in lecture notes.

b. i. By the ideal gas equation, at A:

$pV = NkT$

$$N = \frac{pV}{kT} = \frac{(250\times10^3)\times(7.5\times10^{-3})}{1.38\times10^{-23}\times(623)} = 2.18\times10^{23}$$

ii. Using the idea gas equation, for a fixed amount of gas,

$$pV = nRT \Rightarrow \frac{pV}{T} = \text{const}$$

$$\frac{p_A V_A}{T_A} = \frac{p_C V_C}{T_C}$$

$$V_C = \frac{p_A}{p_C}\times\frac{T_C}{T_A}\times V_A = \frac{250}{101}\times\frac{50}{623}\times(7.5\times10^{-3}) = 1.50\times10^{-3} \text{ m}^3$$

iii. magnitude of work done by the gas = area enclosed by the cycle

$$= \frac{1}{2}\times\left[(250-101)\times10^3\right]\times\left[(7.5-1.4899)\times10^{-3}\right]$$

$= 448$ J

Since the cycle is clockwise, there is a net negative work done <u>on</u> the gas. Thus, the net work done <u>by</u> the gas is positive, and is equal to 448 J.

iv. Zero.

Internal energy only depends on the state of the gas, and after one complete cycle, the gas returns to its initial state, so there is no net change in the gas's internal energy.

3. **[CJC/H2/Prelims 2010/P2/4 (part)]**

a. i. t = 4.0 min, when the gradient of the graph begins to increase

ii. Mass of ice melted in 1.0 minute = gradient of graph

1. When the heater is switched off:

$$\text{gradient} = \frac{(0.011-0.000)}{(4.0-0.0)} = \frac{0.011}{4.0} = 2.75\times10^{-3}\ \text{kg min}^{-1}$$

2. When the heater is switched on:

$$\text{gradient} = \frac{(0.100-0.038)}{(12.6-8.0)} = \frac{0.062}{4.6} = 0.0135\ \text{kg min}^{-1}$$

b. Power produced by the heater: $P = IV = (5.2)\times(11.5) = 59.8$ W

Mass of ice melted in 1.0 minute due to the heater only: $13.48\times10^{-3} - 2.75\times10^{-3} = 10.73\times10^{-3}$ kg s^{-1}

$$P = \left(\frac{dm}{dt}\right)L_{fus} \Rightarrow L_{fus} = \frac{P}{\frac{dm}{dt}} = \frac{59.8}{1.783\times10^{-4}} = 334390 = 334\ \text{kJ kg}^{-1}$$

4. **[Dunman High/H2/Prelims 2010/P3/3]**

a. $Q = mc_{ice}\Delta\theta + mL_{fus}$

$= (0.050)\times(2100)\times(15) + (0.050)\times(3.3\times10^5)$

$= 18100$ J

b. Let m' be the total mass of water in the container.

$$18075 + mc\Delta\theta = m'c\Delta\theta'$$

$$18075 + (0.050)\times(4200)\times(7.5) = m'\times(4200)\times[(30-7.5)]$$

$$m' = \frac{18075 + (0.050)\times(4200)\times(7.5)}{(4200)\times(22.5)}$$

$$m' = 0.208\ \text{kg}$$

c. Lowest temperature is measured when the mixture is in thermal equilibrium.

5. **[IJC/H2/Prelims 2010/P3/2]**

a. i. See definitions in lecture notes.

ii. $\Delta U = Q + W$

ΔU: increase in internal energy of the system

Q: heat supplied to the system

W: work done on the system

b. i. Since there is no change in temperature during boiling, the kinetic energy of the water molecules remains constant.

However, when the volume of the water expands during boiling, the separation between water molecules increases, the potential energy of the atoms increases.

Since internal energy of water is the sum of potential and kinetic energy of the water molecules, there is a change (increase) in internal energy.

ii. The volume occupied by the water molecules increases on vaporisation.

Hence the water molecules have to do work, pushing back the atmosphere.

iii. During boiling, the internal energy of the water molecules increases ($\Delta U > 0$).

According to first law of thermodynamics, the thermal energy supplied must be positive since work done on the boiling water is negative.

6. **[HCI/H2/Prelims 2010/P3/2]**

a. i. See definitions in lecture notes.

ii. See definitions in lecture notes.

b. i. $W_{by\ gas} = p\Delta V = (11\times10^5)(0.070-0.020) = 55$ kJ

ii.

Process	work done on gas, W / kJ	heat supplied to gas, Q / kJ	increase in internal energy, ΔU / kJ
A	+201	**0**	**+201**
B	**−55**	+74	**+19**
C	−185	**0**	**−185**
D	**0**	−35	−35

7. **[MI/H2/Prelims 2010/P2/3 (part)]**

a. i. During melting, molecules move further apart from lattice structure to clusters of molecules.

Energy input is used to gain potential energy without changing kinetic energy of the molecules; hence temperature remains constant.

ii. During melting, intermolecular forces are only partially broken, while during boiling, almost all intermolecular forces are broken since molecules of the liquid are much closer to one another than those of the gas.

Furthermore, the great gain in volume from liquid to gas means that there is a large amount of work done against the atmospheric pressure.

8. **[MJC/H2/Prelims 2010/P3/5]**

a. See definitions in lecture notes.

b. i. Isothermal process, fixed volume of gas is heated and allowed to expand freely

Phase transition, melting of solid / boiling of liquid

ii. Adiabatic process, any processes that are carried out rapidly (so that heat exchange with the environment is negligible), such as pumping air quickly into the bicycle's tyre, or any processes carried out on a well-insulated system, such as a gas expanding in a vacuum flask.

9. **[NJC/H2/Prelims 2010/P3/2]**

a. i. Show calculation to determine value of pV at C, B and one other point on the curve BC

ii. Internal energy of an ideal gas is only made up of the sum of microscopic kinetic energy due to random motion of its molecules.

b. $\dfrac{p_A}{T_A}=\dfrac{p_C}{T_C}$

$$T_A=\frac{p_A}{p_C}T_C=\frac{1.70\times10^5}{3.40\times10^5}\times(385)=192.5\text{ K}=-80.7\text{ °C}$$

c. i.
$$\Delta U_{AB}=\frac{3}{2}nR\Delta T=\frac{3}{2}p\Delta V_{AB}=\frac{3}{2}p(V_B-V_A)=\frac{3}{2}\times(1.7\times10^5)\times[(9.8-5.0)\times10^{-4}]=122\text{ J}$$

ii. W is negative when the gas expands during the change AB.
At the same time, its internal energy has increased (since 120 J > 0).
From the first law of thermodynamics, $\Delta U=Q+W$, energy must be supplied to the system as heat ($Q>0$) and the amount of energy supplied as heat must be greater than the amount of work done ($|Q|>|W|$).

10. [NYJC/H2/Prelims 2010/P3/7]

a. See definitions in lecture notes.

b. i. $T_f=(20+273.15)=293\text{ K}$

ii. $Q=mc\Delta\theta=(0.90)\times(990)\times(180-20)=143\text{ KJ}$

c.
$$p_iV_i=n_iRT_i=\frac{m_i}{M}RT_i$$
$$m_i=\frac{p_iV_iM}{RT_i}=\frac{(1.0\times10^5)\times(0.10)\times(0.030)}{(8.314)\times(180+273.15)}=0.07963\text{ kg}$$
$$p_fV_f=n_fRT_f=\frac{m_f}{M}RT_f$$
$$m_f=\frac{p_fV_fM}{RT_f}=\frac{(1.0\times10^5)\times(0.10)\times(0.030)}{(8.314)\times(25+273.15)}=0.1210\text{ kg}$$
$$\Delta m=m_f-m_i=0.1210-0.07963=0.0414\text{ kg}$$

d.

	Heat supplied to gas / J	Work done on gas / J	Increase in internal energy of gas / J
A to B	0	–300	–300
B to C	–450	0	–450
C to D	0	650	650
D to A	**850**	**–750**	**100**

e. i. This is to ensure that the rate of heat lost from the apparatus to the environment remains unchanged for both of the experiments.

ii.
$$Q=mc\Delta T+Q_{lost}\Rightarrow\frac{dQ}{dt}=\left(\frac{dm}{dt}\right)c\Delta T+\frac{dQ_{lost}}{dt}$$

In the first experiment: $25=\left(\dfrac{0.15}{60}\right)\times c\times(19-15)+P_{lost}$ (1)

In the second experiment: $37=\left(\dfrac{0.23}{60}\right)\times c\times(19-15)+P_{lost}$ (2)

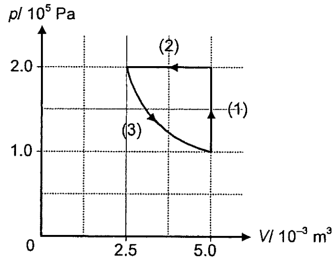
Solving (1) and (2) simultaneously, P_{lost} = 2.5 W

11. [RI/H2/Prelims 2010/P2/4]

a. See definitions in lecture notes.

b. i. $pV=nRT$
$$T=\frac{pV}{nR}=\frac{(1.0\times10^5)\times(5.0\times10^{-3})}{(0.20)\times(8.31)}=301\text{ K}$$

ii.



12. [PJC/H2/Prelims 2010/P3/6]

a. i. $(\Delta p)V=(\Delta n)RT$
$$\Delta n=\frac{(\Delta p)V}{RT}=\frac{[(3.91-2.76)\times10^5]\times(0.0160)}{(8.314)\times(27+273.15)}=0.737\text{ mol}$$

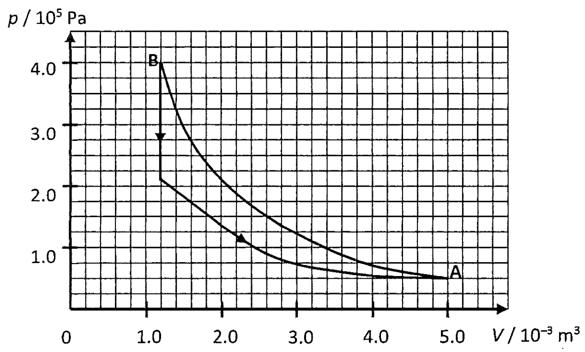
ii.
$$\text{no. of moles of air in supply of air}=\frac{pV}{RT}=\frac{(11.65\times10^5)\times(0.0117)}{(8.314)\times(300.15)}=5.462147\text{ mol}$$
no. of moles of gas needed for 4 tyres $=4\times0.737343=2.94937<5.46$ mol
There is sufficient air in the supply to fill up four tyres.
Pressure in the supply after filling 4 tyres:
$$p_{remaining}=\frac{[(5.462147-2.94937)]\times(8.314)\times(300.15)}{0.0117}=5.36\times10^5\text{ Pa}>4.00\times10^5\text{ Pa (shown)}$$

iii. $\text{Mean KE}=\dfrac{3}{2}kT=\dfrac{3}{2}\times(1.38\times10^{-23})\times(300.15)=6.21\times10^{-21}\text{ J}$

iv. $U=\dfrac{3}{2}NkT=(6.21\times10^{-21})\times(6.023\times10^{23})=3.74\times10^3\text{ J}$

b. i. Using $pV=nRT$, product of p and V is proportional to T.
$p_AV_A=(0.500\times10^5)\times(5.0\times10^{-3})=250\text{ kg m}^2\text{ s}^{-2}$
$p_BV_B=(4.000\times10^5)\times(1.2\times10^{-3})=480\text{ kg m}^2\text{ s}^{-2}$
As $p_BV_B>p_AV_A$, the temperature at B is higher. And so, temperature of air increases during compression.

ii.



13. [SAJC/H2/Prelims 2010/P3/6]

a. Internal energy is the sum of microscopic potential energy and kinetic energy of molecules.

In an ideal gas, there is no potential energy as there are no forces of attraction between molecules.

Hence, internal energy is the sum of microscopic kinetic energy only. Since mean kinetic energy of molecules is proportional to absolute temperature, internal energy is proportional to absolute temperature.

b. minimum kinetic energy of molecules is 0.

Hence, minimum absolute temperature is 0 (or cannot be negative).

14. [TJC/H2/Prelims 2010/P2/4]

a. The increase in internal energy of a system is equal to the sum of the heat transferred to the system and the work done on the system.

15. [YJC/H2/Prelims 2010/P3/2 (part)]

a. Q = thermal energy to raise temperature to boiling point + thermal energy required to vaporise the ethanol.

$$Q = mc\Delta\theta + mL_{vap} = (0.79)\times(1.0)\times[2.4(78-20)+840] = 774 \text{ J}$$

b. During fusion, intermolecular forces are only partially broken, while during boiling, almost all intermolecular forces are broken since molecules of the liquid are much closer to one another than those of the gas.

Furthermore, the great gain in volume from liquid to gas means that there is a large amount of work done against the atmospheric pressure.